

# Quantitative Diagrammatic Axiomatisation of Divergence

Ralph Sarkis (he/him)

joint work with Fabio Zanasi

June 24th, 2026

String Diagrams

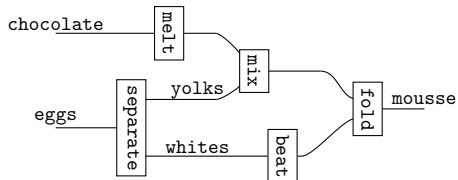
Kullback–Leibler Divergence

Quantitative Diagrammatic Axiomatisation

# Intuition

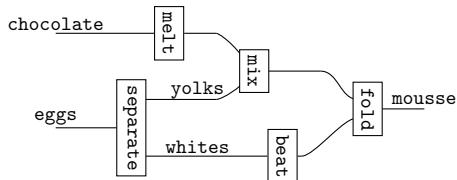
String diagrams are a syntax for processes and systems.

## Example



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## Example


$$\text{fold}(\text{mix}(\text{melt}(\text{chocolate}), \pi_1(\text{separate}(\text{eggs}))), \text{beat}(\pi_2(\text{separate}(\text{eggs})))) \\ (\text{melt} \otimes \text{separate}); (\text{mix} \otimes \text{beat}); \text{fold}$$

# Monoidal Terms

A **generator** is a symbol with an arity  $u$  and a coarity  $v$  in  $\mathbb{N}$ :  $u - \boxed{f} - v$ .

With a set of generators  $\Sigma$ , inductively generate the set of **monoidal terms**  $\mathcal{M}_\Sigma$ .

$$\frac{u - \boxed{f} - v \in \Sigma}{u - \boxed{f} - v \in \mathcal{M}_\Sigma} \text{GEN} \quad \frac{x, y \in \mathbb{N}}{\frac{x}{y} \bowtie \frac{y}{x} \in \mathcal{M}_\Sigma} \text{SWAP} \quad \frac{x \in \mathbb{N}}{x - x \in \mathcal{M}_\Sigma} \text{ID} \quad \frac{}{\dots \in \mathcal{M}_\Sigma} \text{UNIT}$$

$$\frac{u - \boxed{f} - v \quad v - \boxed{g} - w}{u - \boxed{f} - \boxed{g} - w} \text{SEQ}$$

$$\frac{u - \boxed{f} - v \quad u' - \boxed{f'} - v'}{u - \boxed{f} - v \quad u' - \boxed{f'} - v'} \text{PAR}$$

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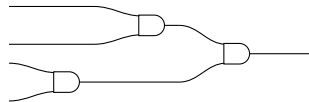
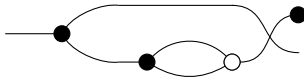
$$\frac{u - \boxed{f} - v \in \Sigma}{u - \boxed{f} - v \in \mathcal{M}_\Sigma} \text{GEN} \quad \frac{x, y \in \mathbb{N}}{\begin{array}{c} x \\ y \end{array} \bowtie \begin{array}{c} y \\ x \end{array} \in \mathcal{M}_\Sigma} \text{SWAP} \quad \frac{x \in \mathbb{N}}{x - x \in \mathcal{M}_\Sigma} \text{ID} \quad \frac{}{\dots \in \mathcal{M}_\Sigma} \text{UNIT}$$

$$\frac{u - \boxed{f} - v \quad v - \boxed{g} - w}{u - \boxed{f} - \boxed{g} - w} \text{SEQ}$$

$$\frac{u - \boxed{f} - v \quad u' - \boxed{f'} - v'}{\begin{array}{c} u - \boxed{f} - v \\ u' - \boxed{f'} - v' \end{array}} \text{PAR}$$

## Examples

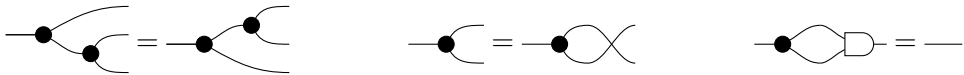
With generators  $\text{---} \bullet \text{---}$ ,  $\text{---} \bullet$ ,  $\text{---} \circ \text{---}$ ,  $\text{---} \sqcup \text{---}$  :





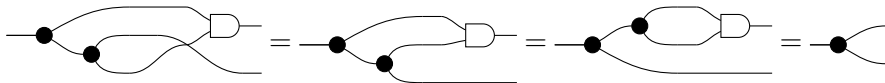
# Equations

We can impose further equations between terms of the same type, e.g.,



Diagrammatic reasoning consists of:

- ▶ Treating string diagrams up to **visually sound manipulations**.
- ▶ Applying **equational reasoning** by replacing diagrams with other diagrams that are known to be equal.





## Definition

Given a set of generators  $\Sigma$  and a set of equations  $E$ : define for any  $u, v \in \mathbb{N}$ ,

$$\mathcal{S}_{\Sigma, E}(u, v) = \{u \text{ --- } \boxed{f} \text{ --- } v \text{ built with } \Sigma\} / \text{diagrammatic reasoning with } E.$$

This freely generates the **syntactic category**  $\mathcal{S}_{\Sigma, E}$ .

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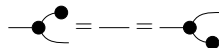
A **presentation** for a category  $\mathbf{C}$  is a pair  $(\Sigma, E)$  such that  $\mathcal{S}_{\Sigma, E} \cong \mathbf{C}$ .

$$\text{Morally, } \frac{\text{string diagrams } u \text{ --- } \boxed{f} \text{ --- } v}{\text{morphisms } f : u \rightarrow v \in \mathbf{C}}$$

## Example 1: Finite Sets and Functions



*Generators*



*Equations*

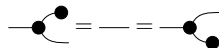
$$\llbracket \text{---} \bullet \text{---} \rrbracket : \{0, 1\} \rightarrow \{0\} = 0 \mapsto 0, 1 \mapsto 0 \qquad \llbracket \text{---} \bullet \rrbracket : \{\} \rightarrow \{0\} = \_ \mapsto$$

*Semantics*

## Example 1: Finite Sets and Functions



*Generators*



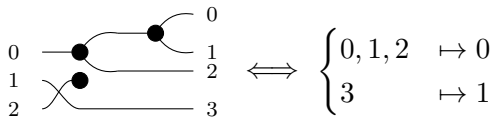
*Equations*

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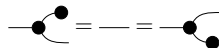
### Example



## Example 1: Finite Sets and Functions



*Generators*



*Equations*

$$\llbracket \text{---} \bullet \rceil \rrbracket : \{0, 1\} \rightarrow \{0\} = 0 \mapsto 0, 1 \mapsto 0$$

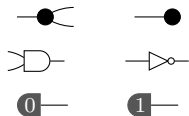
$$\llbracket \text{---} \bullet \rrbracket : \{\} \rightarrow \{0\} = \_ \mapsto$$

*Semantics*

Example

$$\begin{array}{c}
 0 \\
 0 \\
 1 \\
 2
 \end{array}
 \begin{array}{c}
 \text{---} \bullet \\
 \text{---} \bullet \\
 \text{---} \bullet \\
 \text{---} \bullet
 \end{array}
 \begin{array}{c}
 0 \\
 1 \\
 2 \\
 3
 \end{array}
 =
 \begin{array}{c}
 0 \\
 1 \\
 2
 \end{array}
 \begin{array}{c}
 \text{---} \bullet \\
 \text{---} \bullet \\
 \text{---} \bullet
 \end{array}
 \begin{array}{c}
 0 \\
 1 \\
 2 \\
 3
 \end{array}
 \iff
 \begin{cases}
 0, 1, 2 & \mapsto 0 \\
 3 & \mapsto 1
 \end{cases}$$

## Example 2: Boolean Circuits

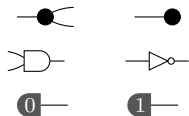


*Generators*

$$\begin{aligned}
 \llbracket \text{CNOT}_{12} \rrbracket &= |x\rangle \mapsto |xx\rangle & \llbracket \text{qubit}_1 \rrbracket &= |x\rangle \mapsto |x\rangle & \llbracket \text{CNOT}_{21} \rrbracket &= |xy\rangle \mapsto |x \wedge y\rangle & \llbracket \text{qubit}_2 \rrbracket &= |x\rangle \mapsto |\neg x\rangle \\
 \llbracket 0 \rrbracket &= |x\rangle \mapsto |0\rangle & \llbracket 1 \rrbracket &= |x\rangle \mapsto |1\rangle
 \end{aligned}$$

*Semantics*

## Example 2: Boolean Circuits



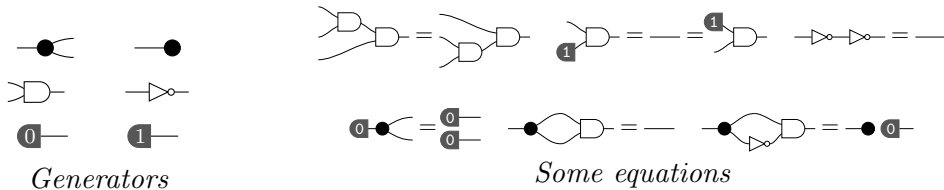
*Generators*

$$\begin{aligned}
 \llbracket \text{fork} \rrbracket &= |x\rangle \mapsto |xx\rangle & \llbracket \text{dot} \rrbracket &= |x\rangle \mapsto | \rangle & \llbracket \text{D-gate} \rrbracket &= |xy\rangle \mapsto |x \wedge y\rangle & \llbracket \text{triangle-gate} \rrbracket &= |x\rangle \mapsto |\neg x\rangle \\
 \llbracket \mathbf{0} \rrbracket &= | \rangle \mapsto |0\rangle & \llbracket \mathbf{1} \rrbracket &= | \rangle \mapsto |1\rangle
 \end{aligned}$$

*Semantics*

$$\frac{u \text{ --- } \boxed{f} \text{ --- } v}{f : \mathbb{B}^u \rightarrow \mathbb{B}^v}$$

## Example 2: Boolean Circuits

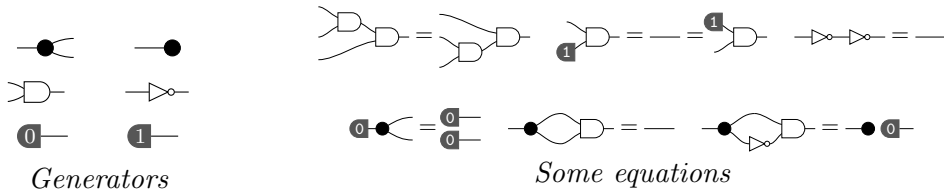


$$\begin{aligned}
 \llbracket \text{---} \bullet \rrbracket &= |x\rangle \mapsto |xx\rangle & \llbracket \text{---} \bullet \rrbracket &= |x\rangle \mapsto | \rangle & \llbracket \text{---} \square \rrbracket &= |xy\rangle \mapsto |x \wedge y\rangle & \llbracket \text{---} \square \rrbracket &= |x\rangle \mapsto |\neg x\rangle \\
 \llbracket \text{---} \square \rrbracket &= | \rangle \mapsto |0\rangle & \llbracket \text{---} \square \rrbracket &= | \rangle \mapsto |1\rangle & & & & \\
 & \text{Semantics} & & & & & & 
 \end{aligned}$$

$$\frac{u \text{---} \boxed{f} \text{---} v}{f : \mathbb{B}^u \rightarrow \mathbb{B}^v}$$



## Example 2: Boolean Circuits



$$\begin{aligned}
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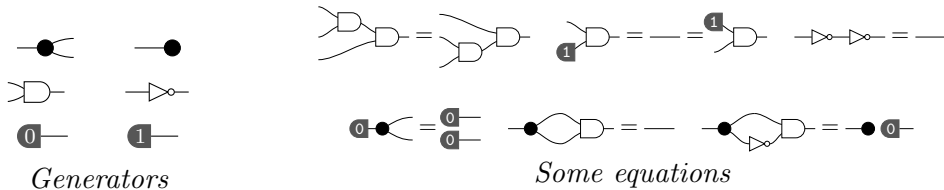
*Semantics*

### Examples

$$\frac{u \text{---} \boxed{f} \text{---} v}{f : \mathbb{B}^u \rightarrow \mathbb{B}^v}$$



## Example 2: Boolean Circuits

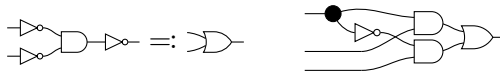


$$\begin{aligned}
 \llbracket \text{---} \bullet \rceil \rrbracket &= |x\rangle \mapsto |xx\rangle & \llbracket \text{---} \bullet \rrbracket &= |x\rangle \mapsto | \rangle & \llbracket \text{---} \text{AND} \rceil \rrbracket &= |xy\rangle \mapsto |x \wedge y\rangle & \llbracket \text{---} \text{NOT} \rceil \rrbracket &= |x\rangle \mapsto |\neg x\rangle \\
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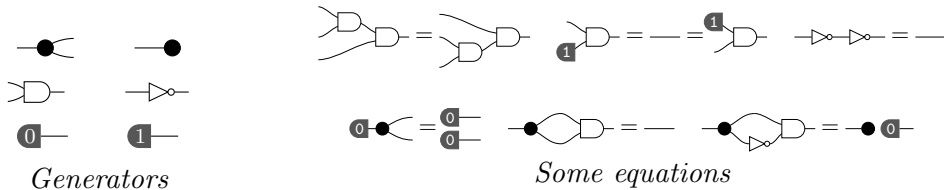
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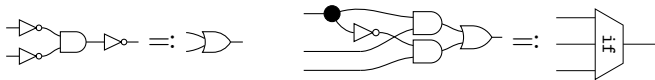


*Semantics*

$$\begin{aligned}
 \llbracket \bullet \rrbracket &= |x\rangle \mapsto |xx\rangle & \llbracket \neg \bullet \rrbracket &= |x\rangle \mapsto |\rangle & \llbracket \neg \neg \bullet \rrbracket &= |x\rangle \mapsto |x\rangle \\
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### Examples

$$\frac{u \text{ --- } \boxed{f} \text{ --- } v}{f : \mathbb{B}^u \rightarrow \mathbb{B}^v}$$



## Example 3: Probabilistic Boolean Circuits

BoolCirc +  
 $\textcircled{p} \text{---}$  ,  $p \in (0, 1)$   
*Generators*

$\llbracket \textcircled{p} \text{---} \rrbracket = |\rangle \mapsto p|1\rangle + (1 - p)|0\rangle$   
*Semantics*

$$\frac{\text{string diagram } u \text{---} \boxed{f} \text{---} v}{\frac{\text{probabilistic map } f : \mathbb{B}^u \rightarrow \mathbb{B}^v}{\text{function } f : \mathbb{B}^u \rightarrow \mathcal{D}(\mathbb{B}^v)}}$$

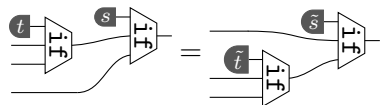
## Example 3: Probabilistic Boolean Circuits

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*Semantics*

BoolCirc +

$$\boxed{p} \text{---} \triangle \text{---} = \boxed{1 - p} \text{---}$$



*Some equations*

$$\frac{\text{string diagram } u \text{---} \boxed{f} \text{---} v}{\frac{\text{probabilistic map } f : \mathbb{B}^u \rightarrow \mathbb{B}^v}{\text{function } f : \mathbb{B}^u \rightarrow \mathcal{D}(\mathbb{B}^v)}}$$

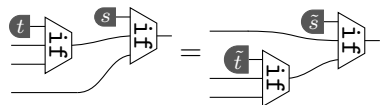
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*Semantics*

BoolCirc +

$$p \text{---} \triangle \text{---} = 1 - p \text{---}$$



*Some equations*

### Examples

string diagram  $u \text{---} \boxed{f} \text{---} v$   


---

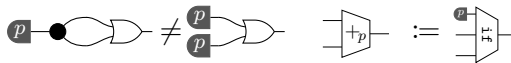
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probabilistic map  $f : \mathbb{B}^u \rightarrow \mathbb{B}^v$   


---

---

function  $f : \mathbb{B}^u \rightarrow \mathcal{D}(\mathbb{B}^v)$



String Diagrams

Kullback–Leibler Divergence

Quantitative Diagrammatic Axiomatisation



Economy · Macro Geopolitics

## Strait of Hormuz traffic returns to normal by end of June?



9% chance ▼ 66%

Polymarket



Strait of Hormuz traffic returns to ...

Yes

Buy Sell

Market ▼

Yes 9¢

No 91¢

Amount

\$0

+\$1

+\$5

+\$10

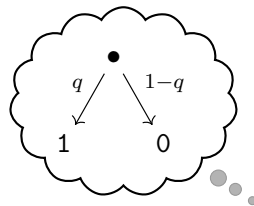
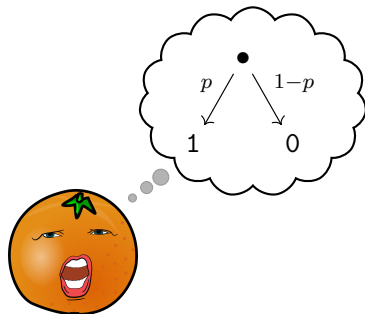
+\$100

Trade

Buy Yes for  $q$ \$ or No for  $(1 - q)$ \$. Reward is 1\$



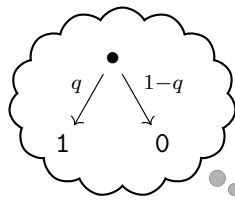
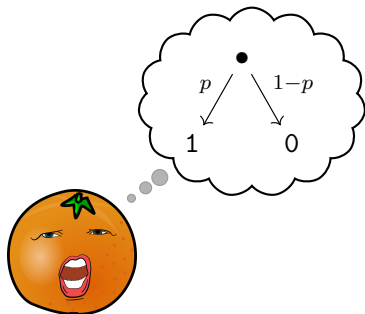
# Betting with Insider Knowledge



► Bad measure

$$\mathbb{E}[\text{gain\_mult}] = p \frac{\text{bet}(1)}{q} + (1-p) \frac{\text{bet}(0)}{1-q}, \text{ bet}(x) \text{ is the proportion of money bet on } x$$

# Betting with Insider Knowledge



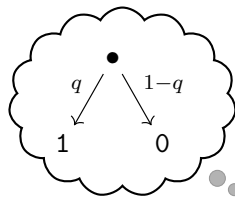
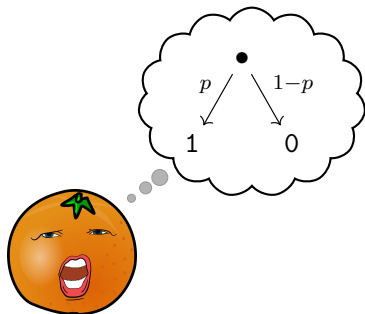
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- Better measure

$$\mathbb{E}[\log(\text{gain\_mult})] = p \log \frac{\text{bet}(1)}{q} + (1-p) \log \frac{\text{bet}(0)}{1-q}$$

# Betting with Insider Knowledge



- ▶ Bad measure

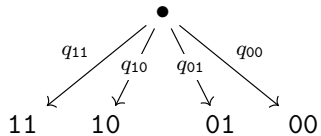
$$\mathbb{E}[\text{gain\_mult}] = p \frac{\text{bet}(1)}{q} + (1-p) \frac{\text{bet}(0)}{1-q}, \text{ bet}(x) \text{ is the proportion of money bet on } x$$

- ▶ Better measure

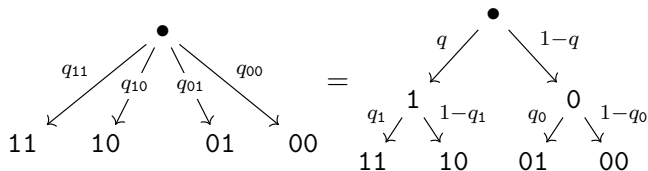
$$\max \mathbb{E}[\log(\text{gain\_mult})] = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q} = \text{kl}\left(\begin{bmatrix} p \\ 1-p \end{bmatrix}, \begin{bmatrix} q \\ 1-q \end{bmatrix}\right)$$

This is the **Kullback–Leibler divergence** between probability distributions.

## Bets with More Outcomes

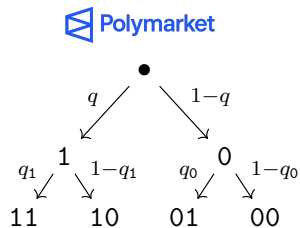
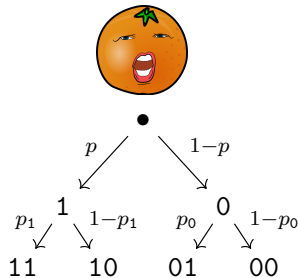


## Bets with More Outcomes



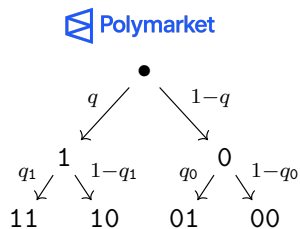
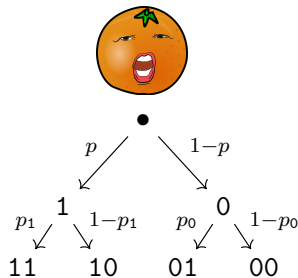
This is the **chain rule** of probability distributions.

# Bets with More Outcomes



$$\max \mathbb{E}[\log(\text{gain\_mult})] = \text{kl}\left(\begin{bmatrix} p \\ 1-p \end{bmatrix}, \begin{bmatrix} q \\ 1-q \end{bmatrix}\right) + p \text{kl}\left(\begin{bmatrix} p_1 \\ 1-p_1 \end{bmatrix}, \begin{bmatrix} q_1 \\ 1-q_1 \end{bmatrix}\right) + (1-p) \text{kl}\left(\begin{bmatrix} p_0 \\ 1-p_0 \end{bmatrix}, \begin{bmatrix} q_0 \\ 1-q_0 \end{bmatrix}\right)$$

# Bets with More Outcomes



$$\text{kl}\left(\begin{bmatrix} p_{11} \\ p_{10} \\ p_{01} \\ p_{00} \end{bmatrix}, \begin{bmatrix} p_{11} \\ p_{10} \\ p_{01} \\ p_{00} \end{bmatrix}\right) = \text{kl}\left(\begin{bmatrix} p \\ 1-p \end{bmatrix}, \begin{bmatrix} q \\ 1-q \end{bmatrix}\right) + p \text{kl}\left(\begin{bmatrix} p_1 \\ 1-p_1 \end{bmatrix}, \begin{bmatrix} q_1 \\ 1-q_1 \end{bmatrix}\right) + (1-p) \text{kl}\left(\begin{bmatrix} p_0 \\ 1-p_0 \end{bmatrix}, \begin{bmatrix} q_0 \\ 1-q_0 \end{bmatrix}\right)$$

This is the **chain rule** of KL divergence.

# Recursive Definition of KL Divergence

Given

- ▶ Let  $\mathbb{B} := \{0, 1\}$  and  $\varphi, \psi$  be distributions on  $\mathbb{B}^n$ ,
- ▶  $p = \sum_y \varphi(1y)$  and  $q = \sum_y \psi(1y)$  are the probabilities that the first bit is 1,
- ▶  $\overline{\varphi}_x$  and  $\overline{\psi}_x$  are the conditional distributions on  $\mathbb{B}^{n-1}$  with the first bit fixed.<sup>1</sup>

The chain rule tells us:

$$\text{kl}(\varphi, \psi) = \text{kl}\left(\begin{bmatrix} p \\ 1-p \end{bmatrix}, \begin{bmatrix} q \\ 1-q \end{bmatrix}\right) + p \text{kl}(\overline{\varphi}_1, \overline{\psi}_1) + (1-p) \text{kl}(\overline{\varphi}_0, \overline{\psi}_0). \quad (1)$$

---

<sup>1</sup> $\overline{\varphi}_1 := y \mapsto \varphi(1y)/p$  and  $\overline{\varphi}_0 := y \mapsto \varphi(0y)/(1-p)$ .

If  $p = 0$  (resp.  $p = 1$ )  $\overline{\varphi}_1$  (resp.  $\overline{\varphi}_0$ ) is assumed to be an arbitrary distribution on  $\mathbb{B}^{n-1}$ .



# Recursive Definition of KL Divergence

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- ▶ Let  $\mathbb{B} := \{0, 1\}$  and  $\varphi, \psi$  be distributions on  $\mathbb{B}^n$ ,
- ▶  $p = \sum_y \varphi(1y)$  and  $q = \sum_y \psi(1y)$  are the probabilities that the first bit is 1,
- ▶  $\overline{\varphi}_x$  and  $\overline{\psi}_x$  are the conditional distributions on  $\mathbb{B}^{n-1}$  with the first bit fixed.<sup>1</sup>

The chain rule tells us:

$$\text{kl}(\varphi, \psi) = \text{kl}\left(\begin{bmatrix} p \\ 1-p \end{bmatrix}, \begin{bmatrix} q \\ 1-q \end{bmatrix}\right) + p \text{kl}(\overline{\varphi}_1, \overline{\psi}_1) + (1-p) \text{kl}(\overline{\varphi}_0, \overline{\psi}_0). \quad (1)$$

This defines  $\text{kl} : \mathcal{D}(\mathbb{B}^n) \times \mathcal{D}(\mathbb{B}^n) \rightarrow [0, \infty]$ , where  $\mathcal{D}$  is the distribution monad.

---

<sup>1</sup> $\overline{\varphi}_1 := y \mapsto \varphi(1y)/p$  and  $\overline{\varphi}_0 := y \mapsto \varphi(0y)/(1-p)$ .

If  $p = 0$  (resp.  $p = 1$ )  $\overline{\varphi}_1$  (resp.  $\overline{\varphi}_0$ ) is assumed to be an arbitrary distribution on  $\mathbb{B}^{n-1}$ .

String Diagrams

Kullback–Leibler Divergence

Quantitative Diagrammatic Axiomatisation

Recall that probabilistic boolean circuits axiomatises  $\mathbf{BStoch}^{\otimes}$ :

- ▶ Objects are natural numbers
- ▶ Morphisms  $n \rightarrow m$  are function  $\mathbb{B}^n \rightarrow \mathcal{D}(\mathbb{B}^m)$

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KL divergence is lifted to homsets  $\mathbf{BStoch}^{\otimes}(n, m)$ :

$$\overline{\text{kl}}(f, g) = \max_{x \in \mathbb{B}^n} \text{kl}(f(x), g(x)).$$

# KL as an Enrichment

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This yields a  $[0, \infty]_+$  **Rel**-enriched category  $\mathbf{BStoch}_{\text{kl}}^{\otimes}$  because

$$\overline{\text{kl}}(f ; g, f' ; g') \leq \overline{\text{kl}}(f, f') + \overline{\text{kl}}(g, g') \quad \overline{\text{kl}}(f \otimes g, f' \otimes g') \leq \overline{\text{kl}}(f, f') + \overline{\text{kl}}(g, g')$$

**Goal:** Find an axiomatisation of  $\mathbf{BStoch}_{\text{kl}}^{\otimes}$ .

# Quantitative Diagrammatic Implications

We use quantitative equations from [MPP16] between string diagrams:

$$u - \boxed{f} - v =_{\varepsilon} u - \boxed{g} - v \Leftrightarrow d(f, g) \leq \varepsilon$$

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[MPP16] Radu Mardare, Prakash Panangaden, and Gordon D. Plotkin. “Quantitative Algebraic Reasoning”. In: *Proceedings of the 31st Annual ACM/IEEE Symposium on Logic in Computer Science, LICS '16, New York, NY, USA, July 5-8, 2016*. Ed. by Martin Grohe, Eric Koskinen, and Natarajan Shankar. ACM, 2016, 700–709. doi: 10.1145/2933575.2934518. URL: <https://doi.org/10.1145/2933575.2934518>

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We generalize that to implications:

$$\frac{-\boxed{f_1}- =_{\varepsilon_1} -\boxed{g_1}- \quad \dots \quad -\boxed{f_n}- =_{\varepsilon_n} -\boxed{g_n}-}{-\boxed{f}- =_{\varepsilon} -\boxed{g}-}$$

## Theorem

*Given a set of generators and a set of quantitative implications, we can generate the free enriched symmetric monoidal category.*

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[MPP16] Radu Mardare, Prakash Panangaden, and Gordon D. Plotkin. “Quantitative Algebraic Reasoning”. In: *Proceedings of the 31st Annual ACM/IEEE Symposium on Logic in Computer Science, LICS '16, New York, NY, USA, July 5-8, 2016*. Ed. by Martin Grohe, Eric Koskinen, and Natarajan Shankar. ACM, 2016, 700–709. doi: 10.1145/2933575.2934518. URL: <https://doi.org/10.1145/2933575.2934518>

# Axiomatising KL Divergence

With the generators and equations of probabilistic boolean circuits, we add:

$$\boxed{f} =_{\varepsilon} \boxed{f'} \quad , \quad \boxed{g} =_{\delta} \boxed{g'} \quad \Rightarrow \quad \boxed{f} \boxed{g} =_{\varepsilon+\delta} \boxed{f'} \boxed{g'}$$

$$\begin{array}{c} \text{---} \boxed{f} \text{---} \\ \text{---} \boxed{f'} \text{---} \end{array} =_{\varepsilon} \begin{array}{c} \text{---} \boxed{g} \text{---} \\ \text{---} \boxed{g'} \text{---} \end{array} \quad , \quad \begin{array}{c} \text{---} \boxed{f'} \text{---} \\ \text{---} \boxed{g'} \text{---} \end{array} =_{\delta} \begin{array}{c} \text{---} \boxed{g'} \text{---} \\ \text{---} \boxed{g} \text{---} \end{array} \Rightarrow \begin{array}{c} \text{---} \boxed{f} \text{---} \\ \text{---} \boxed{f'} \text{---} \end{array} =_{\varepsilon+\delta} \begin{array}{c} \text{---} \boxed{g} \text{---} \\ \text{---} \boxed{g'} \text{---} \end{array}$$

$$\begin{array}{c}
\boxed{f_1} - =_{\varepsilon} \boxed{g_1} - \quad \boxed{f_0} - =_{\delta} \boxed{g_0} - \\
\text{CHAIN}_{\otimes} \\
\begin{array}{c} \boxed{f_1} \\ \boxed{f_0} \end{array} \begin{array}{c} \text{1} \\ \text{0} \end{array} \begin{array}{c} \text{+}_p \end{array} = \text{kl} \left( \begin{bmatrix} p \\ 1-p \end{bmatrix}, \begin{bmatrix} q \\ 1-q \end{bmatrix} \right) + p\varepsilon + (1-p)\delta \quad \begin{array}{c} \boxed{g_1} \\ \boxed{g_0} \end{array} \begin{array}{c} \text{1} \\ \text{0} \end{array} \begin{array}{c} \text{+}_q \end{array} = \\
- \boxed{f_1} - =_{\varepsilon} - \boxed{g_1} - \quad - \boxed{f_0} - =_{\delta} - \boxed{g_0} - \\
\text{IF}_{\max} \\
\begin{array}{c} \boxed{f_1} \\ \boxed{f_0} \end{array} \begin{array}{c} \text{IF}_{\max} \end{array} = \max\{\varepsilon, \delta\} \quad \begin{array}{c} \boxed{g_1} \\ \boxed{g_0} \end{array} \begin{array}{c} \text{IF}_{\max} \end{array} = \\
\overline{f =_0 f} \quad \dots
\end{array}$$



# Axiomatising KL Divergence

With the generators and equations of probabilistic boolean circuits, we add:

$$\boxed{f} =_{\varepsilon} \boxed{f'} \quad , \quad \boxed{g} =_{\delta} \boxed{g'} \quad \Rightarrow \quad \boxed{f} \boxed{g} =_{\varepsilon+\delta} \boxed{f'} \boxed{g'}$$

$$\begin{array}{c} \text{---} \boxed{f} \text{---} \\ \text{---} \boxed{g} \text{---} \end{array} =_{\varepsilon} \begin{array}{c} \text{---} \boxed{f'} \text{---} \\ \text{---} \boxed{g'} \text{---} \end{array} =_{\delta} \begin{array}{c} \text{---} \boxed{f} \text{---} \\ \text{---} \boxed{f'} \text{---} \end{array} =_{\varepsilon+\delta} \begin{array}{c} \text{---} \boxed{g} \text{---} \\ \text{---} \boxed{g'} \text{---} \end{array}$$

$$\begin{array}{c}
\boxed{f_1} - =_{\varepsilon} \boxed{g_1} - \quad \boxed{f_0} - =_{\delta} \boxed{g_0} - \\
\text{CHAIN}_{\otimes} \\
\begin{array}{ccc}
\begin{array}{c} \boxed{f_1} \\ \boxed{f_0} \end{array} \begin{array}{c} \text{1} \\ \text{0} \end{array} \begin{array}{c} \text{+}_p \end{array} = & \text{kl} \left( \begin{bmatrix} p \\ 1-p \end{bmatrix}, \begin{bmatrix} q \\ 1-q \end{bmatrix} \right) + p\varepsilon + (1-p)\delta & \begin{array}{c} \boxed{g_1} \\ \boxed{g_0} \end{array} \begin{array}{c} \text{1} \\ \text{0} \end{array} \begin{array}{c} \text{+}_q \end{array} = \\
\boxed{f_1} - =_{\varepsilon} \boxed{g_1} - \quad \boxed{f_0} - =_{\delta} \boxed{g_0} - & \text{IF}_{\max} & \overline{f =_0 f} \quad \dots
\end{array}
\end{array}$$

## Theorem

*This theory axiomatises  $\mathbf{BStoch}_{\text{kl}}^{\otimes}$ .*

# Future Work

## More Examples

- ▶ *Already done*: KL divergence, Rényi divergence, total variation
- ▶ *Looking into*: Jensen–Shannon divergence, quantum processes
- ▶ *Dream*: distance for evaluating probabilistic learning models

## Quantitative Program Logics

- ▶ Diagrammatic Hoare logics with tape diagrams in [BDGDL25]
- ▶ Quantitative Hoare logics in many papers.
- ▶ *Dream*: Quantitative diagrammatic Hoare logics

## Extend the Framework

- ▶ *Fibred/Doubled*: to account for distances between in/outputs (Kantorovich)
- ▶ *Graded*: for distances that depend on auxiliary information.

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[BDGDL25] Filippo Bonchi, Alessandro Di Giorgio, and Elena Di Lavore. “A diagrammatic algebra for program logics”. In: *Foundations of software science and computation structures*. Vol. 15691. Lecture Notes in Comput. Sci. Springer, Cham, 2025, pp. 308–330. ISBN: 978-3-031-90896-5; 978-3-031-90897-2. DOI: 10.1007/978-3-031-90897-2\_15. URL: [https://doi.org/10.1007/978-3-031-90897-2\\_15](https://doi.org/10.1007/978-3-031-90897-2_15)

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